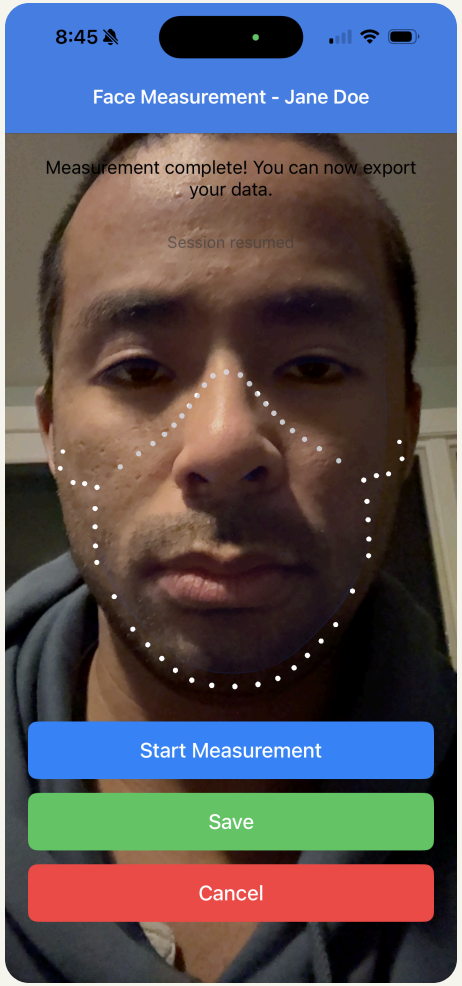
From face capture to useful measurements.

Lessons from building MasqFit and Breathesafe

Edderic Ugaddan · Breathesafe LLC

For NPPTL researchers developing iOS and Android apps

ARKit in MasqFit: five facial paths.



ARKit in the current app



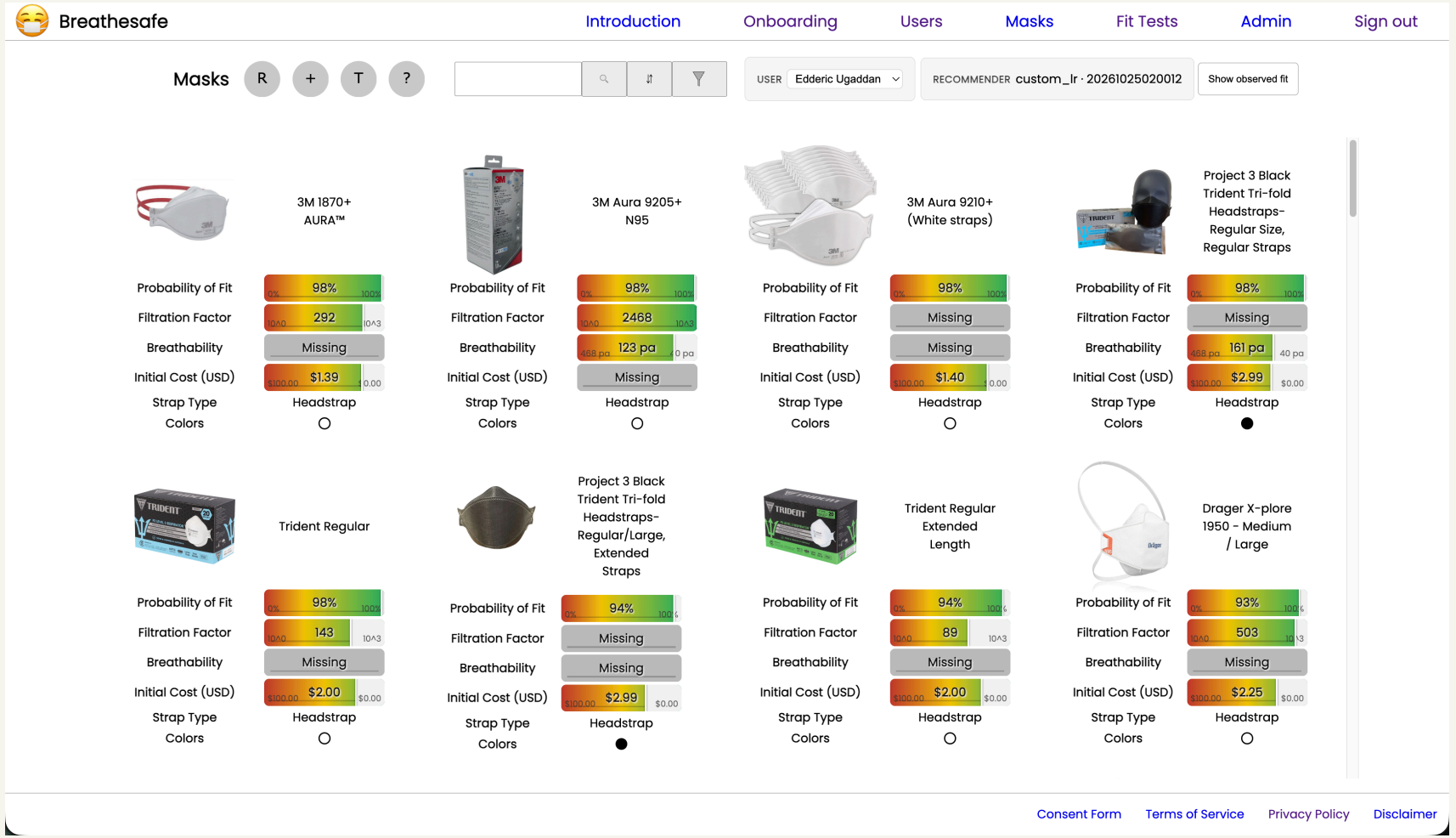
Five paths in millimeters

Five measured paths

Nose · Strap · Top cheek
Mid cheek · Chin

Nose, top cheek, mid cheek, and chin form the perimeter proxy used for ranking.

What the model recommends for my face.



October 29 screenshot · October 25 custom_lr model · Observed Fit hidden

Presenter screenshot · Individual scores are model outputs; calibration is not established.

Keep the nose path separate from total perimeter.

$$\text{New logit} = \text{perimeter-model logit} - \lambda \times [(N_{face} - \mu_{mask/style}) / 10]^2$$

Observed face nose path + learned mask nose preference · $\lambda > 0$ · lengths in mm

Held-out metric	Perimeter deviation	+ Nose deviation
Log loss ↓	1.108	**0.931**
Precision ↑	56.4%	**63.9%**
Recall ↑	58.0%	**59.4%**
F1 ↑	0.572	**0.615**

1,086 tests · 49 people · fixed 50% cutoff · 5 user-grouped folds · mean of 3 initializations

> Improved on mixed measured/imputed inputs; the gain did not reproduce with measured-only training.

A stronger shortlist, with a Top-1 tradeoff.

CV hit rate	Perimeter deviation	+ Nose deviation	Change
Top-1	54.2%	53.0%	**-1.2 pp**
Top-3	79.5%	81.3%	**+1.8 pp**
Top-5	82.0%	85.2%	**+3.2 pp**

> Top-3 and Top-5 improve; the first-ranked result is slightly worse.

Ranks previously tested observations for the 32 people with ≥1 positive fit label.
Retrospective relevance; prospective success with newly recommended masks remains untested.

The tradeoff: metric scale, speed, and reach.

Approach	Geometry & scale	Experience	Main constraint
ARKit + TrueDepth Our iOS path	Metric 3D mesh; meters → millimeters	Real-time	Apple hardware path; anatomical mapping still needed
Google MediaPipe	Normalized image coordinates + relative depth; physical scale needs calibration	Real-time	A landmark mesh is not a subject-specific metric measurement
Photogrammetry	Multi-view 3D surface; physical scale must be established	~45 s raw reconstruction Earlier developer report	Slower reconstruction; anatomical landmarking is a separate step

> Standard-camera capture offers a path toward iOS, Android, and webcams.

Apple geometry · Google coordinate conventions · Timing: presenter report, not a controlled comparison.

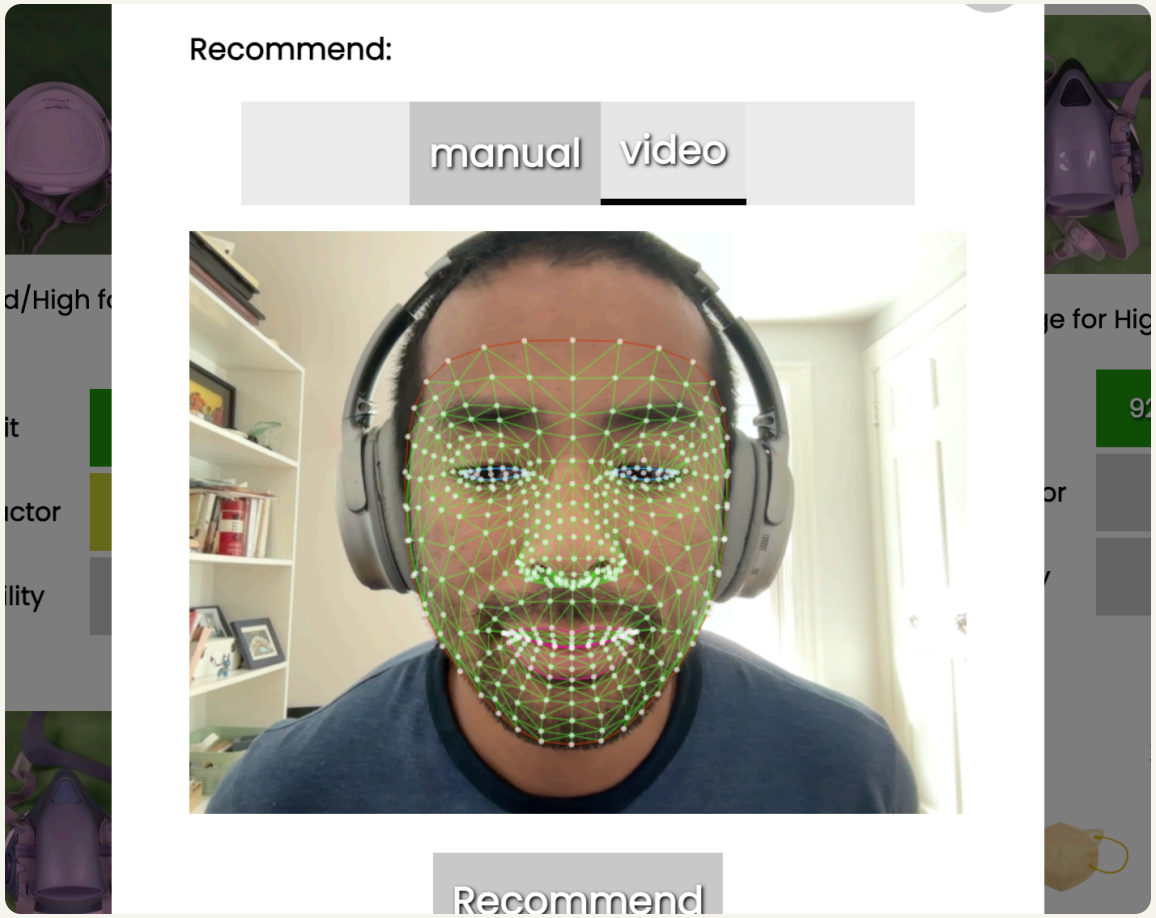
ARKit vs. caliper: an exploratory self-check.

Real-time metric output made capture easy. Agreement varied across six measurements on my face.

Measurement	ARKit (mm)	Caliper (mm)	|Difference| (mm)	Relative difference*
Face width	142	137.0	5.0	3.6%
Face length	125	112.0	13.0	11.6%
Nose breadth	36	30.0	6.0	20.0%
Lower face length	74	64.7	9.3	14.4%
Nose protrusion	16	14.0	2.0	14.3%
Lip width	47	52.0	5.0	9.6%

* Absolute difference ÷ caliper value. One person, one comparison; causes of the differences have not been isolated.

A mesh vertex is not an anatomical definition.



MediaPipe mesh in a MasqFit prototype

Anatomical correspondence

Mesh vertices may not coincide with the traditional landmarks used by NIOSH.

Occluded points

Face models can infer hidden points. An estimated coordinate is not a direct observation.

> Open question: how accurate are hidden points inferred from one viewing angle?

Google: predicted mesh · Apple: fitted face geometry · Single-angle accuracy: not yet established by this project.

NEW / CAMERA PROTOTYPE · 0:40

Photos now produce a physically scaled digital headform.



Raw photogrammetry reconstruction



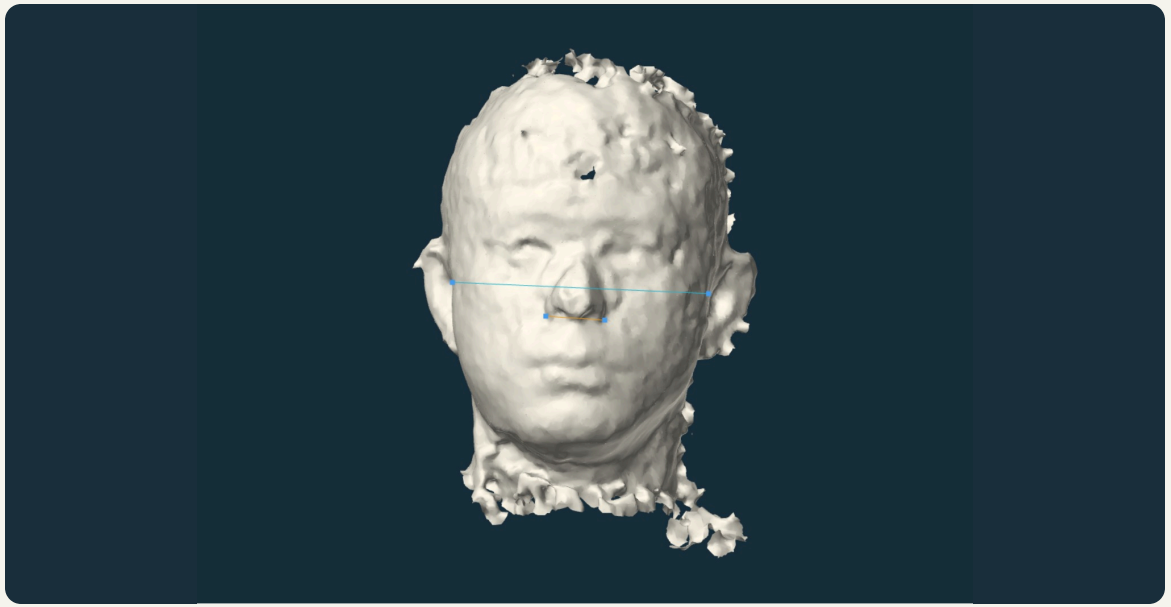
Smoothed surface · same coordinates

61 ordinary-camera photos → registered 3D surface → supported smoothing → tape-based scale in millimeters.

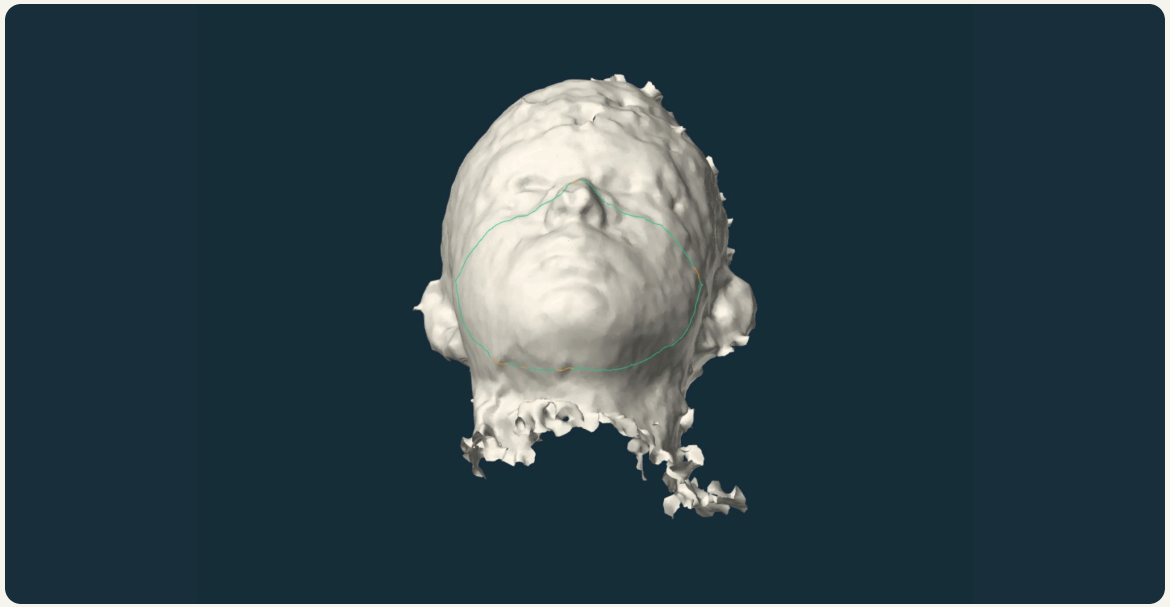
Head still, camera moving; eye-level and lower views retain upper-neck coverage. Laptop prototype demonstrated; Android workflow validation remains ahead.

NEW / LANDMARKS & DISTANCES · 0:40

Landmarks support chords and surface routes.



Straight chords · nose and cheekbone widths



Surface-route draft · inspect below the jaw

Dense multi-photo correspondences propose surface-attached points. Original-photo overlays support anatomical review.

A chord joins two 3D endpoints. A surface arc follows the mesh through controls. Named landmarks and route controls retain their attachments.

Actual prototype views · Width contacts visually approved for this capture · Route overlays may show through the mesh; no seal-pressure prediction.

NEW / INITIAL PHYSICAL CHECKS · 0:40

Three linear checks agree within 2.3 mm in one pilot.

Scale reference: 284 mm Median bitragion–subnasale tape arc: 288, 280, 284 mm.

Dimension	Caliper median	Headform	Headform − caliper
Nose width	35.0 mm	35.2 mm	+0.2 mm
Face length	114.6 mm	112.3 mm	-2.3 mm
Face width	150.9 mm	153.0 mm	+2.1 mm

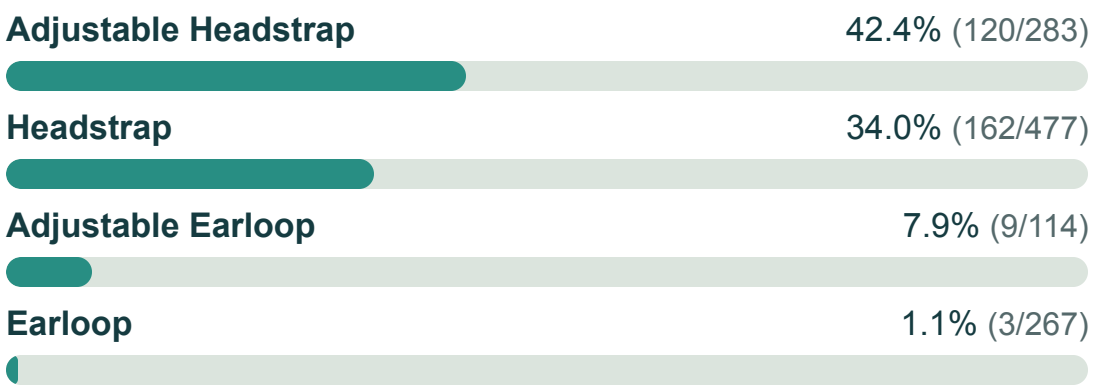
> Caliper dimensions check the tape-derived scale; they do not adjust it.

One person, one capture; three caliper readings per dimension. Method and endpoint choices were inspected during development—not a blinded accuracy or capture-repeatability study.

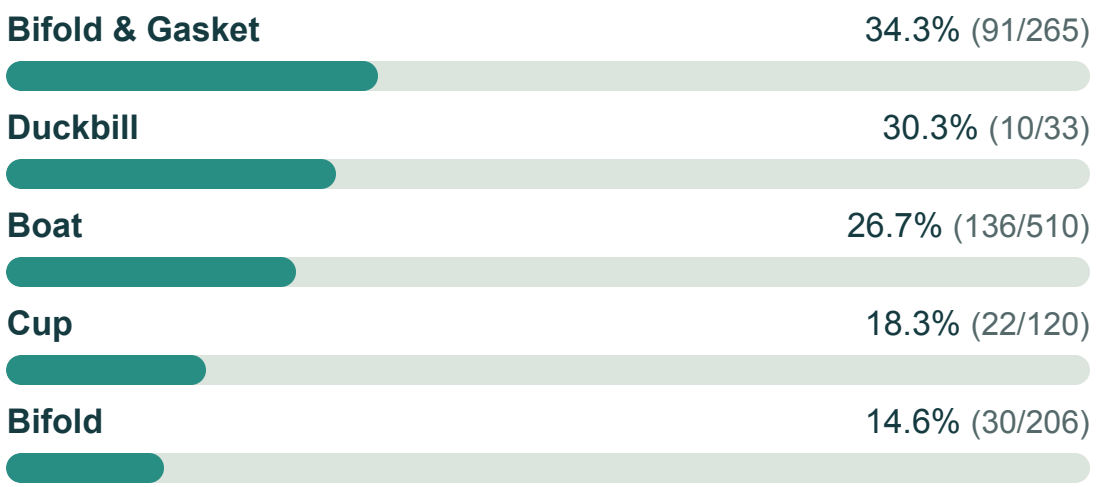
Observed pass rates vary by strap and style.

296 of 1,145 tests passed 25.9% overall in the dashboard's filtered dataset

Strap type



Mask style



Descriptive test-level rates; multiple tests can come from one person. Groups with fewer than 10 tests omitted.

Who contributed the 1,145 fit tests?

55 unique participants · counts below are people, not tests

Gender / sex

Cisgender male	22
Cisgender female	26
Prefer not to say	7

Race / ethnicity

White / Caucasian	41
Asian / Pacific Islander	6
Multiple ethnicity / Other	5
Prefer not to say	3

Age

26-35	6
36-45	9
46-55	6
56-65	5
Prefer not to say	29

Representation and repeat testing matter when interpreting a dashboard rate.

Breathesafe dashboard snapshot Oct 29, 2026 · Small groups may be pooled into “Prefer not to say.”

A complementary scope for respirator research.

MasqFit and Breathesafe add community-use questions alongside NIOSH's facial-measurement work.

01

More mask standards

Catalog options include KN95 and FFP2/FFP3, alongside NIOSH-approved respirators.

02

Children included

Contributed data include children and child-sized masks. Pediatric prediction accuracy still needs evaluation.

03

Open methods

GPLv3 code lets researchers inspect, run, and extend methods—supporting reproducible science.

Shared opportunity: comparable measurements and fit outcomes across platforms and populations.

NIOSH face-measuring app description · CDC on child-sized respirators · Breathesafe GPLv3 source.

Acknowledgements.

Wayne

Community fit testing and training

Mish

Initial mask recommendations and Masks for Kids

Lisa Foreman

Community fit testing and the Readimask Dental Nose Hack

Gerard Hughes

Breathability and fit-testing data

Amanda Abbott

PDX Respirator Repository

Nukit

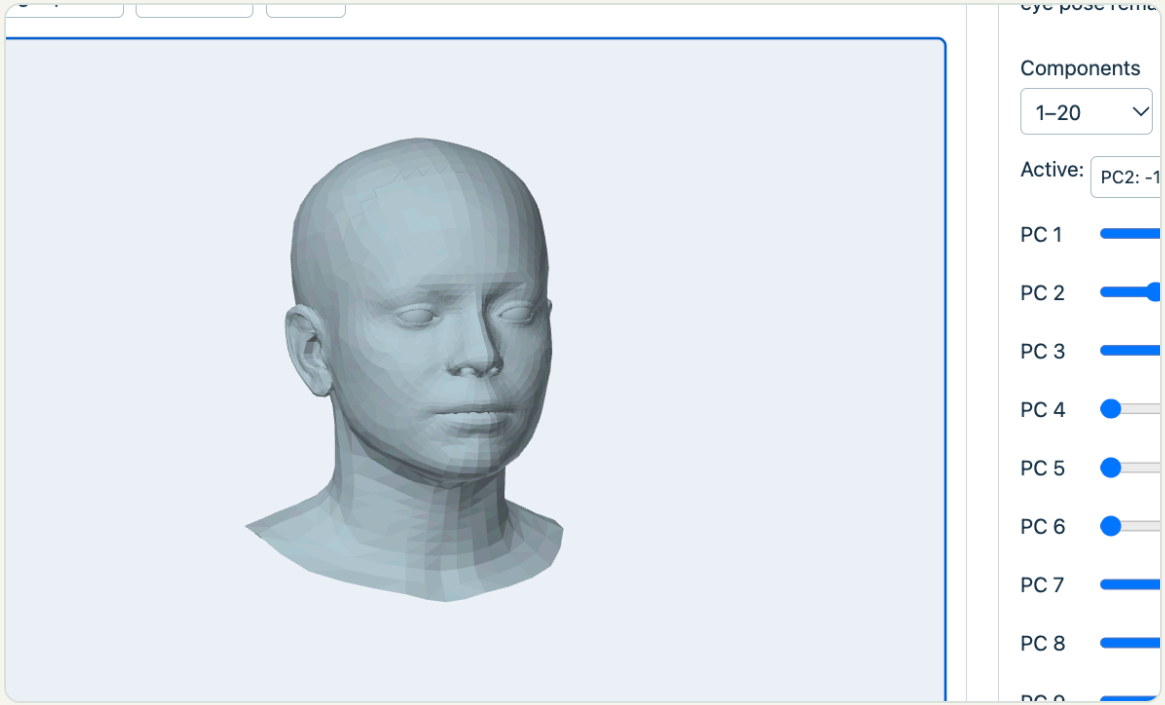
Fit-testing probes

With thanks to Kanro for early support, and to Aaron Collins / OpenAeros, research participants, and anonymous contributors.

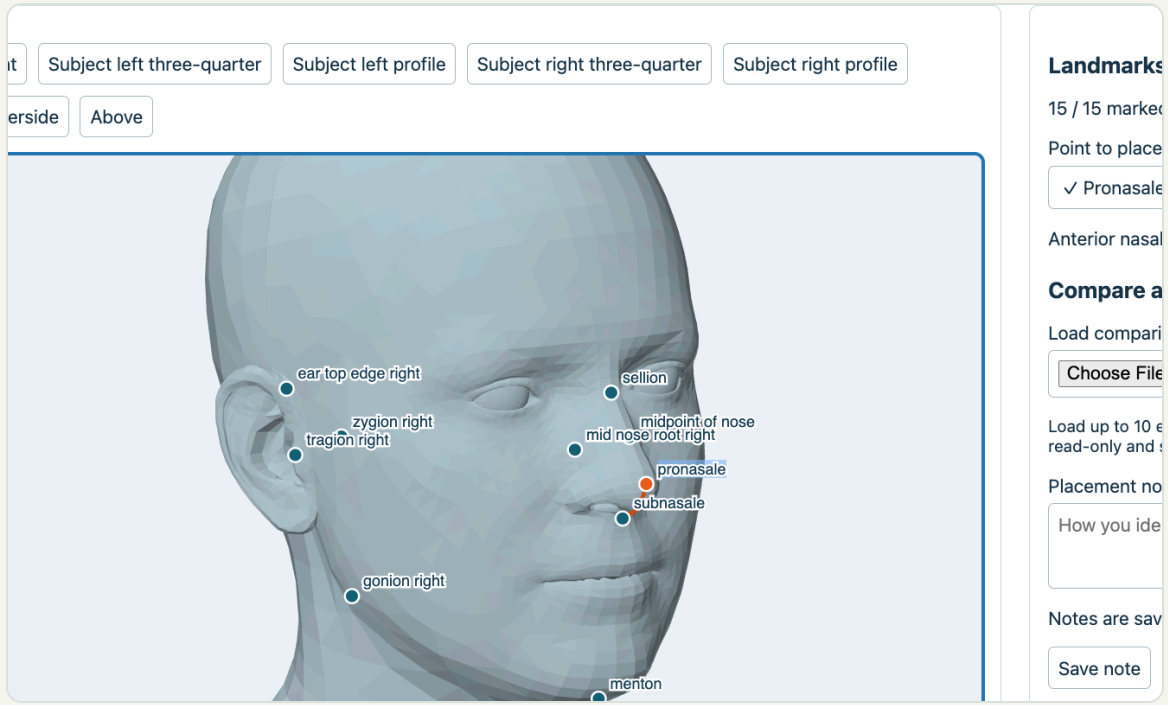
Seeking funding to continue this research · Open to work and consulting

info@breathesafe.xyz · LinkedIn: Edderic Ugaddan

Synthetic faces remain a complementary research path.



Generate controlled 3D faces



Label anatomical points on the 3D surface

3D face → 2D views → **CNN detector · planned** eyes · nose · ears · other landmarks → 3D projection → dimensions → fit

October research-tool screenshots · Complementary synthetic-landmark work; not the current prototype output or Android app screens.

The prototype uses one physical scale anchor.

Absolute scale

The prototype uses a 284 mm bitragion–subnasale tape arc to establish scale.

A canonical average face is a model prior, not an independent ruler for this person.

Surface correspondence

Use the matching camera intrinsics and pose to project a 2D detection onto the mesh.

Check occlusion, multiple intersections, missing surface regions, and agreement across views.

> Evaluate absolute millimeter errors without fitting scale to the reference target.

Working prototype scale: 284 mm / 4.28220 reconstruction units ≈ 66.3211 mm per unit. Caliper checks do not change scale.

Sources and boundaries of the claims.

Technical references

Apple: ARFaceGeometry

Apple: face tracking configuration and device support

Google: Face Mesh coordinate systems and canonical model

Photogrammetry evidence

Aldridge et al. (2005), Digital three-dimensional photogrammetry: evaluation of anthropometric precision and accuracy using a Genex 3D camera system

Evidence concerns the tested system and protocol; it does not validate our camera-only pipeline.

Earlier ~45-second raw reconstruction is presenter-reported, not a full-workflow benchmark. The new one-capture caliper checks are not a blinded or multi-person accuracy study.